Syllabus - CSCI 2670

Course CSCI 2670 (Theory of Computation)
Semester Spring, 2026
Prerequisite CSCI 2610 (Discrete Mathematics)
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Office hours

Announcements

This is the general syllabus for CSCI 2670. The following guidelines give a general description of the course and related policies. This syllabus may be updated with new or different information. You are responsible for any announcements about the class that are made in class or posted to eLC. The updated version of this syllabus will be posted to the eLC class page.

Course description

The theory of computing, including finite automata, regular expressions and languages, context-free grammars and languages, push-down automata, pumping lemmas, the Chomsky hierarchy of language classes, Turing machines and computability, undecidability of the halting problem, reducibilities among decision problems and languages, time and space complexity, and NP-completeness and tractability.This is a first course on the theory of computing.

Contacting me

You may email me at the address above. Your emails should be sent from your UGA address and should Include the course name and section number. Other emails may not get a reply. If your email is not addressed, then you may remind me or ask me in class. Do not message me on eLC; I cannot reply to eLC messages. If you have questions about grades, then please email the TAs and CC me.

Class materials

Textbook Theory of automata and formal languages, by Bill Hollingsworth

This textbook is only available in the Grafstate Margin Notes. You will need a Grafstate account (https://grafstate.com). Grafstate is a program that I wrote and we will use it for lessons and assignments.

Grafstate

I use Grafstate to give lectures instead of using slides. It can be used as a note taking tool as well as a simulator to help you visualize difficult and abstract concepts.

Instructions for working with Grafstate are available from the Grafstate website. Development is ongoing and I try to respond to needs and issues that arise in class or with assignments. If you discover a bug in the software, then please email me and tell me how to encounter the bug. If you have suggestions or feedback, then please tell me.

Learning outcomes

By the end of the course, students should be able to:

1. Define regular languages using various methods and be able to prove if a language is or is not regular.
2. Define context-free languages using various methods and be able to prove if a language is or is not context-free.
3. Define semidecidable languages using various types of Turing machines and be able to prove the different types of machines are equivalent with regard to the languages they recognize.
4. Define decidability and be able to prove if a language is decidable or not.
5. Distinguish between P and NP and prove a language is in P or in NP.

Topical Outline

1. Regular languages
   a. Given an NFA M, create a DFA or regular expression that recognizes L(M).
   b. Given a regular language L, create an NFA that recognizes L.
   c. Use the pumping lemma for regular languages to prove that a language is not regular.
2. Context-free languages
   a. Given a description of a context-free language L, develop a context-free grammar G such that L(G) = L.
   b. Convert a context-free grammar to an equivalent pushdown automaton and vice-versa.
   c. Convert a CFG into Chomsky normal form (CNF).
   d. Given a context-free grammar G in CNF and a string w, use the CYK algorithm to determine if G generates w
   e. Use the pumping lemma for context-free languages to prove that a language is not context-free.
   f. Identify if a given language is regular, context-free, or neither.
3. Semidecidable languages
   a. Given a language L, create a Turing machine that recognizes L.
   b. Convert between the different variations of Turing machines (e.g., multitape to single- tape).
   c. Create a Turing machine that computes a function.
4. Decidability
   a. Define decidability and determine if a language is decidable.
   b. Prove that the Halting problem is undecidable.
   c. Reduce one problem to another.
   d. Use reductions to prove that a problem is undecidable.
5. Computational Complexity
   a. Define P, NP, and NP-complete
   b. Show a problem is in P.
   c. Write pseudo-code describing a nondeterministic Turing machine's steps to solve a problem.
   d. Describe the Cook-Levin theorem.
   e. Describe a verifier for an NP-complete language.

Attendance and notes

Attendance is mandatory. You are expected to come to class and take notes. I do not plan to post my lecture notes. Coming to class, taking notes, and asking questions are important for being successful in the course.

Attendance will be checked each Monday. If you are not present when the TA is there to check, then you will not be counted present. If you are not present on Monday, then you may not work in a group with others on the CW for that week. Attendance will be checked periodically on Tuesdays and Thursdays. Your attendance score will be computed as the number of times present divided by the number of checks. This score will count as part of the CW portion of the grade.

Assignments

Except when stated otherwise, all assignments must be typed using Grafstate.

Late policy

Classwork and homework assignments may be turned in late with the following constraints:

1. 10% is deducted per day. In other words, an assignment handed in 2 days late cannot receive higher than 80%.
2. Assignments more than 2 days late will not be accepted.

It is your responsibility to check that you receive a grade for each assignment. If you believe that a grade is incorrect or missing, then please email the TAs (with a CC to me) within 1 week of the grades being posted requesting an explanation or further consideration. After the 1-week period, the grades for that assignment may not be revisited.

Grading scale

I use the following grading scale:

Letter	Range
A	93-100
A-	90-92
B+	87-89
B	83-85
B-	80-82
C+	77-79
C	73-75
C-	70-72
D	60-69
F	0-59

Making up assignments

You may make up an assignment that you miss due to an excused absence. For an absence to be excused, you must show signed documentation of the absence within 3 days of the absence. If the absence is due to illness, then the documentation must be signed by a doctor on letterhead. Scheduling something that conflicts with class is not considered an excused absence.

If an absence is excused for an exam, then a make-up exam may be offered. If the absence is not excused, then a grade of 0 will be given.

Grading weights

Grades will be weighted as follows:

Type	Weight
Classwork and homework	30%
Midterm exams	20% each
Final exam	30%

Exams

The dates for the final exam are:

Section	Date	Time	Place
61727	Fri., Dec. 5	3:30 - 6:30 pm	Monday room
22423	Wed., Dec. 10	3:30 â€“ 6:30 pm	Monday room

The dates for the midterm exams are:

Exam	Date
Midterm 1	Thu., Sep 18
Midterm 2	Thu., Oct 16

- You must come to the exam for your section.

Academic honesty

Unless explicitly allowed, artificial intelligence-based technologies, such as ChatGPT, must not be used to generate responses for student assignments.

You are responsible for knowing and abiding by the following policies:

- A Culture of Honesty, the University's policy and procedures for handling cases of suspected dishonesty - www.uga.edu/ovpi.
- School of Computing - Policy Statement on Academic Honesty.

Other information

Class Accommodation

Students with a disability or health-related issue who need a class accommodation should make an appointment to speak with the instructor as soon as possible. Students who require such an accommodation for an exam must contact the instructor at least two weeks before the exam date.

Mental Health and Wellness Resources

If you or someone you know needs assistance, you are encouraged to contact Student Care and Outreach in the Division of Student Affairs at 706-542-7774 or visit https://sco.uga.edu/ . They will help you navigate any difficult circumstances you may be facing by connecting you with the appropriate resources or services.

UGA has several resources for a student seeking mental health services (https://www.uhs.uga.edu/bewelluga/bewelluga) or crisis support (https://www.uhs.uga.edu/info/emergencies).

If you need help managing stress anxiety, relationships, etc., please visit BeWellUGA (https://www.uhs.uga.edu/bewelluga/bewelluga) for a list of FREE workshops, classes, mentoring, and health coaching led by licensed clinicians and health educators in the University Health Center.

Additional resources can be accessed through the UGA App.